

Mathematics A

Unit 1 • Chapter OL-T04 Classified

Cross-session • chapter-organised candidate

6 classified items • 10 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Unit 1 • Chapter OL-T04 • 6 items • 10 marks

Chapter	Items	Marks	Starts
01. OL-T04 Powers, Roots & Standard Form	6	10	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Powers, Roots & Standard Form

Unit 1 • 10 marks • 6 item(s)

November 2023 | 4MA1-2H Q12 | 2 marks

Power of a power or solve a numerical index relation

June 2023 | 4MA1-1H Q15 | 3 marks

Power of a power or solve a numerical index relation

November 2021 | 4MA1-1H Q1 | 1 marks

Multiply or divide powers with a common base

November 2021 | 4MA1-2H Q1 | 1 marks

Multiply or divide powers with a common base

November 2021 | 4MA1-2H Q1 | 1 marks

Power of a power or solve a numerical index relation

November 2021 | 4MA1-2H Q1 | 2 marks

Power of a power or solve a numerical index relation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 24

4MA1-2H Q12 (a)

ORIGINAL QUESTION UNIT 1

2 marks

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$$x = \dots\dots\dots$$

(2)

Worked solution 24

Solve an index equation

MODULAR UNIT 1

2 marks

(a) – Solve an index equation

Write all terms in base 2

$$2^7 \cdot 4^5 = 2^7 \cdot (2^2)^5 = 2^{17}, \quad 4^x = (2^2)^x = 2^{2x}.$$

Equate exponents

$$2x = 17 \Rightarrow x = 8.5.$$

Final answer

$$x = 8.5$$

Question 08

4MA1-1H Q15

ORIGINAL QUESTION

3 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$
(3)

Worked Solution 08

Numerical indices

MODULAR UNIT 1

3 marks

Write every term as a power of 2

$$\sqrt{2} = 2^{1/2}, \quad 8^3 = 2^9, \quad 16^{3/2} = 2^6.$$

Subtract indices

$$2^{1/2} \div 2^{9-6} = 2^{1/2-3} = 2^{-5/2}.$$

Final answer

$$n = -\frac{5}{2} = -2.5$$

1 (a) Simplify $e^8 \div e^2$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions • Variant: Multiply or divide powers with a common base

Method / working

1. (a) $(e^8 \div e^2 = e^6)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (a) e^6

1 (a) Write down the value of m , given that $3^4 \times 3^5 = 3^m$

$m = \dots\dots\dots$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions • Variant: Multiply or divide powers with a common base

Method / working

1. (a) $(3^4 \times 3^5 = 3^9)$, so $(m=9)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $m=9$, $n=21$, $p=4$

(b) Write down the value of n , given that $(5^3)^7 = 5^n$

$n = \dots\dots\dots$

(1)

Worked solution

November 2021 | Question 1 | 1 marks

Family: Simplify numerical index expressions • Variant: Power of a power or solve a numerical index relation

Method / working

1. (b) $((5^3)^7 = 5^{21})$, so $(n=21)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $m=9$, $n=21$, $p=4$

(c) Find the value of p , given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

$p = \dots\dots\dots$
(2)

Worked solution

November 2021 | Question 1 | 2 marks

Family: Simplify numerical index expressions • Variant: Power of a power or solve a numerical index relation

Method / working

1. (c)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $\frac{7^8 \times 7^2}{7^p} = 7^{10-p} = 7^6$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. So $(10-p=6)$, hence $(p=4)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. Answer: $(m=9)$, $(n=21)$, $(p=4)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $m=9$, $n=21$, $p=4$